

26 Naive Bayes Classification

We have seen how to fit Bayesian regression models for predicting numerical variables. Now we will introduce a Bayesian *classification* method for predicting *categorical* variables. The goal is to “classify” observations according to which category (class) they belong to.

This handout corresponds to Chapter 14 of (Johnson et al. 2022).

There exist multiple penguin species throughout Antarctica, including the *Adelie* (A), *Chinstrap* (C), and *Gentoo* (G). Our goal will be to classify penguins according to their species given characteristics like weight or bill length. (Throughout “penguin” refers to an Antarctic penguin of one of these three species.)

Example 26.1. Suppose first that we have no information about a randomly selected penguin (other than it’s an Antarctic penguin from one of these three species).

1. How might we formulate a prior distribution for the species of the penguin? Would we necessarily assume equal prior probability for the three species?
2. Suppose that among Antarctic penguins of these three species, 44.2% of Adelie, 19.8% are Chinstrap, and 36.0% are Gentoo. Start to set up a Bayes table.
3. Given any randomly selected penguin, in the absence of any information about the penguin, what specify would you predict it to be? Why? What is the probability that your classification is correct?

Example 26.2. Now suppose we know that a randomly selected penguin is below average weight (that is, below 4200g).

1. What is the next column of the Bayes table to fill in? What information would we need to do so?
2. Suppose that 83.4% of Adelie penguins are below average weight, 89.7% of Chinstrap penguins are below average weight, and 4.9% of Gentoo penguins are below average weight. Among these three species, the likelihood of being below average weight is greatest for Chinstrap. Does that mean we should necessarily classify the penguin to be Chinstrap? Why?
3. Complete the Bayes table and find the posterior probability of each species given that the penguin is below average weight.
4. Given any randomly selected penguin that is below average weight, what specify would you predict it to be? Why? What is the probability that your classification is correct?
5. Now suppose the the penguin is not below average weight. Compute the posterior probability of each species given that the penguin is not below average weight.

6. Given any randomly selected penguin that is not below average weight, what specify would you predict it to be? Why? What is the probability that your classification is correct?

7. Now suppose that given any randomly selected penguin you observed whether or not the penguin is below average weight and then classify it according to the rule you determined above. What is the posterior probability that you classify the penguin correctly?

Example 26.3. Now suppose we know that a randomly selected penguin has a bill length of 50mm. We'll start by assuming we don't know yet whether or not it is below average weight.

1. Start to create a Bayes table. What is the prior? What is the next column of the Bayes table to fill in? What information would we need to do so?

2. Suppose that bill lengths (mm) follow a $N(38.8, 2.66)$ distribution for Adelie, a $N(48.8, 3.34)$ distribution for Chinstrap, and a $N(47.5, 3.08)$ distribution for Gentoo. How would you fill in the likelihood column?

3. Among these three species, the likelihood of having of a 50mm bill is greatest for Chinstrap. Does that mean we should necessarily classify the penguin to be Chinstrap? Why?

4. Complete the Bayes table and find the posterior probability of each species given that the penguin has a 50mm bill.

5. If the species has a 50mm bill, what species would you classify it as? Why?

6. Now suppose before measuring bill length we had already known that the penguin was below average weight. How would our Bayes table have changed? If we don't change the likelihood column, what are we assuming?

7. Compute the posterior probability of each species given that the penguin is below average weight with a 50mm bill, assuming conditional independence between weight and bill length. Given a species with these characteristics, what species would you classify it as? Why?

- **Naive Bayes classification** uses Bayes rule to predict the class of a categorical response variable y given predictor variables $x_1, \dots, x_p$
- Predictor variables can be categorical or numerical
- Naive Bayes classification assumes that predictors are *conditionally independent* given each class

- Naive Bayes classification typically assumes that any numerical predictor variable is conditionally Normal
- Given observed data
 - Prior probabilities for the response variable are estimated to be the observed sample proportion for each class
 - For a categorical predictor x , the likelihood of value $\tilde{x}$ is estimated to be the observed sample proportion of $\tilde{x}$ within each class
 - For a numerical predictor x , the likelihood is computed assuming that within each class values of the predictor follow a Normal distribution with mean and SD estimated to be the sample mean and SD within each class.

Example 26.4. Now suppose we know that a penguin has a bill length of 50mm and a flipper length of 195mm. (We'll ignore whether or not it is below average weight for this example.)

1. Starting with our original prior, what is the “evidence” that the likelihood is based on? What information do we need to fill in the likelihood column?
2. Suppose that flipper lengths (mm) follow a $N(190, 6.54)$ distribution for Adelie, a $N(196, 7.13)$ distribution for Chinstrap, and a $N(217, 6.48)$ distribution for Gentoo. (Recall that bill lengths (mm) follow a $N(38.8, 2.66)$ distribution for Adelie, a $N(48.8, 3.34)$ distribution for Chinstrap, and a $N(47.5, 3.08)$ distribution for Gentoo.) How would you fill in the likelihood column? What are we assuming?
3. Complete the Bayes table and find the posterior probability of each species given that the penguin has a 50mm bill and a 195mm flipper.
4. If the species has a 50mm bill and a 195mm flipper, what species would you classify it as? Why?